

ADITHYA BHASKAR

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EDUCATION

2023-28	Ph.D. in Computer Science, Princeton University, USA <i>Advised by Prof. Danqi Chen</i> (Specialization : Natural Language Processing) Research interests : LLM post-training, LLM tool-use, Mechanistic Interpretability	4.00/4.00
2019-23	Bachelor of Technology in Computer Science and Engineering (Honors), IIT Bombay, India	9.67/10.00
2017-19	High School, Central Board of Secondary Education, India	97.2/100.0
2017	Senior Secondary School, Central Board of Secondary Education, India	98.0/100.0

PUBLICATIONS

2025	Unintentional Unalignment : Likelihood Displacement in Direct Preference Optimization, Noam Razin, Sadhika Malladi, Adithya Bhaskar , Danqi Chen, Sanjeev Arora, and Boris Hanin	ICLR 2025
2024	Finding Transformer Circuits With Edge Pruning, Adithya Bhaskar , Alexander Wettig, Dan Friedman, and Danqi Chen	NeurIPS 2024 (Spotlight)
2024	The Heuristic Core : Understanding Subnetwork Generalization in Pretrained Language Models, Adithya Bhaskar , Dan Friedman, and Danqi Chen	ACL 2024 (Oral)
2023	Benchmarking and Improving Text-to-SQL Generation under Ambiguity, Adithya Bhaskar* , Tushar Tomar*, Ashutosh Sathe, and Sunita Sarawagi	EMNLP 2023
2023	Prompted Opinion Summarization with GPT-3.5, Adithya Bhaskar , Alexander R. Fabbri, and Greg Durrett	ACL 2023 (Findings)
2023	Performance Bounds for LASSO under Multiplicative Noise : Applications to Pooled RT-PCR Testing, Richeek Das, Aaron Jerry Ninan, Adithya Bhaskar , and Ajit Rajwade	Signal Processing, Vol. 214

PREPRINTS

2025	Language Models That Think, Chat Better, Adithya Bhaskar* , Xi Ye*, and Danqi Chen	arXiv preprint, arXiv:2509.20357
2025	Cache Me If You Can : How Many KV's Do You Need for Effective Long-Context LMs?, Adithya Bhaskar* , Alexander Wettig*, Tianyu Gao, Yihe Dong, and Danqi Chen	arXiv preprint, arXiv:2506.17121
2024	Continual Memorization of Factoids in Language Models, Howard Chen, Jiayi Geng, Adithya Bhaskar , Dan Friedman, and Danqi Chen	arXiv preprint, arXiv:2411.07175
2024	Improving Language Understanding from Screenshots, Tianyu Gao, Zirui Wang, Adithya Bhaskar , and Danqi Chen	arXiv preprint arXiv:2402.14073

INVITED TALKS

April 2024	The Heuristic Core : Understanding Subnetwork Generalization in Pretrained Language Models <i>Host : Mathew Monfort</i>	Amazon AWS
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SCHOLASTIC ACHIEVEMENTS

2024	Recipient of the Hisashi and Masae Kobayashi *67 Fellowship .
2023	Recipient of the Thomas Dooie Class of 1974 Research Award .
2019	All India Rank 18 in JEE Advanced 2019 among 240 thousand candidates.
2019	All India Rank 114 in JEE Mains 2019 among 1.1 million candidates.
2018	Secured a position in the top 39 ranks in the Indian National Physics Olympiad and was invited to the Orientation-cum-Selection-Camp in Physics held in May-June 2018.
2018	Secured a position in the top 49 ranks in the Indian National Chemistry Olympiad and was invited to the Orientation-cum-Selection-Camp in Chemistry held in May-June 2018.
2016	Among the 39 students to clear the Indian National Mathematical Olympiad , becoming one of the youngest to ever be invited to the Orientation-cum-Selection-Camp in Mathematics aged 14 .

EXPERIENCE

UT Austin Summer 2022	Research Intern, NATURAL LANGUAGE PROCESSING, USA Advisor : Prof. Greg Durrett Very Large Language Models for Multi-Document Summarization > Developed metrics to measure factuality, faithfulness and specificity (whether it is correct, prefers major viewpoints, and is not too generic) for a summary of multi-document text such as hotel reviews. > Utilized the above along with an n-gram abstractiveness metric to benchmark GPT-3.5, and showed that simple hierarchical summarization of large text performs poorly on faithfulness and specificity. > Investigated various pre-clustering and pre-summarization methods and illustrated that pre-summarization with a pretrained keyword-based extractive model improves correctness, faithfulness and specificity, while only marginally affecting abstractiveness.
Uppsala University Summer 2021	Research Intern, FORMAL VERIFICATION, Advisor : Prof. Parosh Abdulla Model Checking for Programs Running under the ARMv8 Memory Model > Developed a model and simulator for programs running under the ARMv8 memory model, and demonstrated the equivalence of the model to the ARM specification using 7500+ litmus tests. > Used Context Bounded Model Checking to perform State Reachability Analysis for programs under the ARMv8 memory model, achieving up to an order of magnitude of speedup over existing checkers.

OTHER PROJECTS

Robust Models Spring 2023	Bachelor's Project, NATURAL LANGUAGE PROCESSING, Guide : Prof. Sunita Sarawagi Automated data augmentation for robustness. > Demonstrated that training a Text-to-SQL model on partially masked (underspecified) inputs leads to diversity in the model outputs, including in columns/tables, string literals, integers, and aggregates. > Filtered the outputs by model probabilities relative to the output with the unmasked question. > Furnished questions for the generated queries via an SQL-to-Text model. Data augmentation with the pairs led to increases in accuracy on the SPIDER dataset and the robustness benchmark, Dr. SPIDER. > The increases exceeded those obtained by augmenting with Dr. SPIDER style perturbations.
Group Testing Fall 2022	R&D Project, COMPRESSIVE SENSING, Guide : Prof. Ajit Rajwade Applying compressive sensing to improve COVID-19 Group Testing. > Proved the theoretical applicability of Compressive Sensing with Weighted LASSO for any general noise model with an asymptotically well-defined Moment-Generating-Function. > Derived values of the optimal weights for the case of Multiplicative Gaussian Noise as in RT-PCR tests. > Demonstrated improvements in sensitivity, specificity, MCC and RMSE by Monte Carlo Simulations.
C Decompiler Fall 2020	Course Project, SOFTWARE SYSTEMS, Guide : Prof. Amitabha Sanyal Recovering Code From Compiled RTL > Built a decompiler to convert Register Transfer Language to C for portability across architectures. > Utilized lex and bison to parse source code in RTL and identify program elements like assignments, basic arithmetic operations, conditional/looping constructs, function calls and memory accesses. > Performed local & global data flow analysis and control flow analysis to contextualize parsed code.

TEACHING

Spring 2025	Graduate Teaching Assistant, <i>COS 484 : Natural Language Processing</i> Instructors : Danqi Chen, Vikram Ramaswamy, and Tri Dao	Princeton University
Fall 2024	Graduate Teaching Assistant, <i>COS 597R : Deep Dive into Large Language Models</i> Instructors : Danqi Chen and Sanjeev Arora	Princeton University

SERVICE

2025	NeurIPS 2025	Reviewer
2025	ICML 2025 MOSS Workshop	Reviewer
2024	NeurIPS 2024 ATTRIB Workshop	Reviewer